Supplementary information

Repairing manipulation programs under wrist range restrictions

Author-maintained supplementary methods and results.

Reading map. Methods S1–S6; Can outcomes S7–S8; Lift S9; access and numerical reproduction S10–S11; source differences and recorded scenes S12; model selection S13. Section identifiers retain the main text's cross-references.

S1 Source Programs and Exact Input Conventions

The public HDF5 demonstration supplies actions, the full initial state, model XML and environment arguments. Normal replay restores that state without physical projection. The controller reference/cache is initialized to the original arm state under the common current-reference protocol; arm and gripper guards are shared. This changes controller initialization, not the physical initial state.

Each action runs for 25 native physical steps at 20 Hz. The source program takes goal_pos[::25] and goal_ori[::25] from the controller trace and the original last action component as the gripper command. Thus p_t, R_t are desired absolute targets, not measured end-effector poses. The initial reference q^0 is one seven-vector, distinct from the normal joint trajectory used by clipping. Normal traces separately determine qualification, the fixed gate and original carry window. Edited commands execute from the original initial state without stage resets.

For the first desired pose p_0, R_0, initial observed object position o, and $r_t = R_0^\top R_t$, the features are

$$\rho_t = (r_{00}, r_{01}, r_{10}, r_{11}, r_{20}, r_{21}), \qquad \tau_t = t/(N-1), \qquad s_t = [(p_t - p_0)R_0/(0.3\,\mathrm{m}); \rho_t; g_t; \tau_t].$$

Positions use row-vector multiplication. The first two rotation columns are flattened in NumPy C order, not column-by-column. The 11-D source feature, 14-D initial context and 14-D capability feature use

$$c = [q^0/\pi; (o - p_0)R_0/(0.3\,\mathrm{m}); (o_w, o_x, o_y, o_z)], \qquad \tilde{b} = (l_1, u_1, \ldots, l_7, u_7)/\pi.$$

The object quaternion retains observed WXYZ order; it is not rotated into the source-relative position frame. Features and query time are float32 in the network. Generated latents are converted to float64 before materialization against original source arrays, preserving exact zero-edit commands. Padding is masked in attention and loss. The network does not receive full normal traces, target outcomes or future restricted states; geometry, clipping and teacher construction use privileged simulator FK, while neural forward does not.

Finite rigid poses, bounds and references and a unit observed quaternion are required. Materialization requires $l_j + 0.02 < q_j^0 < u_j - 0.02$ for all seven arm joints. The middle-carry taper requires $B - A \geq 4$ actions and unchanged window anchors. Invalid inputs are rejected without projecting the physical state or shortening carry. These pre-existing conditions introduce no new exclusions in the reported scored sets.

S2 Geometric Solution-to-Command Mapping

For selected bounded solution q_t^{sol} at an active source action,

$$q_t^{\mathrm{geom}} = q^0 + w_t(q_t^{\mathrm{sol}} - q^0), \tag{S1}$$

$$R_t^{\mathrm{geom}} = R_t \exp\big([w_t \log(R_t^\top R_{\mathrm{FK}}(q_t^{\mathrm{sol}}))]_\times\big). \tag{S2}$$

Here log returns a rotation vector. Zero gate restores source orientation and q^0; XYZ, gripper and timing are copied. These are controller commands, not physical state assignments. Geometry gates reference displacement; learning gates latents before the nonlinear bounded map. Accepted teacher targets are already materialized and are not gated a second time.

S3 Geometric and Training Implementation Settings

Geometry. SciPy 1.15.3 least_squares minimizes half the squared residual norm of main Eq. (4), with default trust-region-reflective solver and two-point finite-difference Jacobian. Old/position-priority/free variants use $\lambda_R = 0.05/0.005/0$. Each start permits 40 solver evaluations and at most 400 FK calls including numerical derivatives and final evaluation; ftol, xtol and gtol are 10^{-8}. Only optimization seeds are projected inside inset bounds.

An active segment is a maximal run with $w_t > 0$; zero gate clears the preceding solution. Normal posture is q[25t], not q^0. At segment start, regularization uses bounded normal posture; starts are normal posture, the same normal posture again, and bounds midpoint. Subsequently, the first start/regularizer uses the previous solution, with normal posture and midpoint retained. Candidates are ranked by the same objective and start-order ties, never restricted outcomes. A false optimizer success flag does not reject a finite bounded candidate with matching recomputed cost. Invalid values, cost mismatch or excess FK budget cause errors, not infeasibility certificates. The duplicated initial start is part of the retained recipe.

Network. The model has 184,202 parameters: width 64, four heads, two pre-LN encoder and two decoder layers, final layer norms, FFN width 128 and zero dropout. Separate bounds/context MLPs are $14 \to 64 \to 64$; the time MLP is $4 \to 64 \to 64$, with intermediate GELU and final layer norms. Bounds/context embeddings are appended to memory and added to queries from $\phi(\tau) = (\tau, \tau^2, \sin \pi\tau, \cos \pi\tau)$. Both heads are zero-initialized. Attention is noncausal over the known source program. Main Fig. 1 describes this learned branch, not the full comparison system; its photograph is a scene preview, not RGB input.

Composed training. gated_training_prediction precedes imitation_loss: rotation/reference latents are multiplied by the fixed source gate, then real bounds define the reference through $\tanh(w_t z_t)$. This is not equivalent to gating a decoded displacement. Teacher rotations/references are materialized and execution-accepted. Rotation loss compares relative rotation matrices; common left multiplication by source rotation preserves the Frobenius norm. Padding and adjacent-pair masks exclude invalid positions. Main Eq. (5)'s final term penalizes reference-error differences, not physical velocity. Can uses AdamW, batch 16, learning rate 3×10^{-4}, weight decay 10^{-4} and global gradient clipping at 1. Sampling with replacement assigns probability 0.5 to each repair/identity stratum. Expansion resumes model, optimizer and RNG states at update 2,000 for 3,000 further updates.

Constant control. Only the pre-MLP capability feature is replaced by the frozen original-36-TRAIN mean. Initial context, real bounds, initial reference and ROM-dependent gate remain; true bounds still define the output map and loss normalization. The same recorded vector is retained in all expanded Constant models, without recomputing it from TRAIN tensors. Its entries and checkpoint hashes are in protocols/selected_models_and_constant.json. This tests the added explicit token, not removal of all capability information. All six final models use the selected 5,000-update family; the complete sequential selection history is in S13.

S4 Exact Retrieval Recipe

Squared Euclidean distance is applied to

$$d = [\mathrm{vec}(S_{16})/\sqrt{176}; c/\sqrt{14}; \tilde{b}/\sqrt{14}; \log(N/128)],$$

where S_{16} linearly resamples the 11 source features to 16 normalized-time anchors. Preprocessing is as in S1, and ties choose the first frozen bank entry. The bank contains accepted TRAIN programs, not query outcomes.

For a materialized teacher, $v_t^* = \log(R_t^\top R_t^*)$ and $a_{tj} = (q_{tj}^* - q_j^0)/\Delta_j$, where Δ_j is the appropriate upper/lower inset span from q_j^0. Set $z_{tj}^* = \mathrm{arctanh}(a_{tj})$ for $|a_{tj}| < 1$, or $20\,\mathrm{sign}(a_{tj})$ at the boundary; excess beyond one by more than 10^{-12} is rejected. At positive teacher gates, store $(v_t^*/w_t, z_t^*/w_t)$ without an extra small-gate cutoff. Zero gates store zeros and require zero teacher edits. Selected templates are linearly resampled in normalized time, multiplied by the query gate and materialized with its source and true bounds. An exact raw-input/gate match returns the accepted cached program. Both banks use the same descriptor and reconstruction.

S5 Clipping, Control and Telemetry

Native goal thresholds are unchanged: Can requires an in-bin object and $1 - \tanh(10\|p_{\mathrm{eef}} - p_{\mathrm{obj}}\|) < 0.6$; Lift requires cube height above table offset plus 0.04 m. Neither endpoint goal certifies carry or joint legality.

Clipping. At each action boundary, $q_t^c = \mathrm{clip}(q_t^{\mathrm{src}}, l + \epsilon, u - \epsilon)$. Source-to-FK orientation is interpolated on SO(3) by the gate, and references are $q^0 + w_t(q_t^c - q^0)$. Common clipping uses w_t, guard-only b_t; XYZ/gripper remain unchanged. Neither assigns physical state. The gate records

$$a^0 = M^{-1}\{\mathrm{clip}(\tau^0, \tau_{\min}, \tau_{\max}) - b\}$$

as filter_nominal_acceleration, before torque correction, not simulator qacc. Main Eq. (3) is checked at all 25 J7 samples/action; violations beyond 10^{-10} mark an action. Five-action triangular padding gives b_t, and the unchanged carry taper gives $w_t = \max(b_t, h_t)$. Gates are fixed before restricted execution.

Execution order. Absolute pose, original gripper command and posture reference are installed at each action boundary. Native OSC computes nominal torque; the arm filter records a^0 and returns a qualified correction or raw nominal fallback; physics advances without state assignment. Range is checked at both boundaries of every physical step, carry before and after each original carry action, and native goal at the prescribed endpoint. Fifty zero-edit checks reproduce source-copy outcomes, failures included.

All 650 Can edits and 300 Lift jobs retain the same six-axis OSC_POSE constructor: $k_p = 150$, damping ratio 1, fixed impedance, decoupled position/orientation, no pose limits and no interpolation. Constructor control_delta=true and action scaling apply to raw replay; edited set_goal receives explicit absolute pose once per action, checked against its registered goal. Reference changes call update_initial_joints before applying that pose; unchanged references skip this update.

Gripper guard. For native normalized targets x, let s_i be motor-target half ranges. Variant arm_open2 uses open/closed insets $o = 0.002$, $c = 0.0002$ m, and clips x between

$$(-1 + c/s_1, -1 + o/s_2) \quad \text{and} \quad (1 - o/s_1, 1 - c/s_2).$$

The native motor ramp remains. Motor-target protection neither projects finger positions nor guarantees legal execution.

Arm correction. Main Eq. (3) supplies acceleration bounds for all seven joints with 0.02 rad inset. If a^0 is feasible, raw nominal torque is returned. Otherwise, first clip acceleration to the box; accept only if reconstructed torque obeys motor bounds and residual checks. If motor bounds fail, SLSQP minimizes $\|a - a^0\|^2/2$ under both constraint sets (100 iterations, ftol= 10^{-10}), followed by torque clipping/rechecking. Active slack is $\le 10^{-7}$; NNLS multipliers permit 200 iterations. Both correction paths require feasibility residual $\le 10^{-8}$, stationarity/complementarity $\le 10^{-5}$ and zero multiplier negativity. An unqualified correction returns raw nominal torque, not a finding of infeasibility. The seven-joint approximation does not certify contacts, passive forces or coupling. Inspection listings remain in methods/fixed_control_v538.

Carry starts at the first action grasped before and after, ending before the next negative raw gripper command or at source length; every action must pass both checks. Missing/broken windows fail qualification without shortening. Grasp uses native _check_grasp with gripper/target contact geometries. The plotted all-nine margin is

$$m_k = \min_{j=1,\dots,9;\, s\in\{\text{before,after}\}} \frac{\min(q^s_{k,j} - l_j, u_j - q^s_{k,j})}{u_j - l_j}.$$

This dimensionless display does not replace the 10^{-9} execution tolerance.

S6 Teacher Construction Coverage

The Can TRAIN bank's 20 sources count contributors, not all attempted nominations. The earlier 36 eligible conditions yielded 31 repairs and five identities from 108 geometry programs (105 new, three reused); pre-qualification nomination totals remain unverified.

Table S1. Can teacher flow. Source–ROM conditions are correlated; qualification exclusions and missing teachers remain distinct.

Stage	Count	Meaning
Expansion nominations	36	12 sources, three fitted ROMs
Normal-qualification exclusions	6	demo_185 and demo_102
Eligible expansion conditions	30	No replacement of excluded sources
No accepted teacher	7	demo_117/medium; all ROMs of 142 and 156
New accepted teachers	23	18 repairs, five identities
Combined bank	59	Earlier 36 plus new 23; 49 repairs, ten identities

These TRAIN selections do not change FINAL's 50 scored conditions and ten normal-qualification NA conditions.

S7 Can Outcomes, Two Clocks and Earlier Cohorts

S7.1 Source-level uncertainty

The exploratory bootstrap resamples ten whole sources with their five ROMs, methods and matched seeds (10,000 draws; seed 20261008). Full–Constant uses

$$d_g = \frac{1}{15}\sum_{k=0}^{2}\sum_{r=1}^{5}(S^{\text{Full}}_{g,r,k} - S^{\text{Constant}}_{g,r,k}), \qquad \widehat{\delta} = \frac{1}{10}\sum_{g=1}^{10} d_g.$$

Full–clip substitutes clipping for Constant. Percentiles use NumPy 1.26.4 linear interpolation. Fixed models exclude retraining uncertainty; 150 seed–condition outcomes are not independent scenes. This post-protocol analysis is not preregistered.

Full–Constant is positive in four sources, negative in three and zero in three: mean 8.00 percentage points, 95% interval $[-6.00, +23.33]$. Full–clip is 15.33 points, interval $[-4.67, +37.33]$. Both cross zero, leaving population superiority and token benefit unresolved, not establishing equivalence. Figure S5 shows every source difference.

S7.2 Criterion components and additional-action clock

Among 38 original-clock source failures, 12 fail goal/carry only, nine fail all three checks, 16 fail goal only and one fails range only. Finger violations occur in nine conditions and a wrist violation in one. Every paired condition records guard intervention and a state/actuator change; these are not unique evidence of wrist contact or IK infeasibility. Goal/carry/range marginals overlap and are not a causal decomposition.

Table S2. Can criterion components. Strict is the intersection of goal/carry/range. Additional-action repair/regression denominators are 32/18, not original-clock 38/12.

	Original clock /50				+40 clock		
Method	Goal	Carry	Range	Strict	Strict /50	Repair /32	Regress. /18
Source copy	13	29	40	12	18	0	0
Clip / common	50	45	40	36	36	23	5
Clip / guard	50	45	40	36	36	23	5
Nearest / 36	39	38	41	37	37	19	0
Nearest / 59	36	35	39	33	33	15	0
Geometry / old	50	46	50	46	46	28	0
Geometry / position	50	47	49	47	47	29	0
Geometry / free R	47	43	50	43	43	25	0
Full / seed 0	49	48	48	45	45	28	1
Full / seed 1	47	45	45	43	43	26	1
Full / seed 2	49	47	47	43	44	26	0
Constant / seed 0	47	46	45	45	45	27	0
Constant / seed 1	41	42	45	37	38	20	0
Constant / seed 2	39	44	41	37	41	23	0

Figures S1–S2 retain all 840 cells per clock: 700 scored plus 140 NA, including excluded sources 188/170, in registered order. ROMs within a source are correlated. At +40, source-copy rises from 12 to 18 successes: its own 32/18 failure/success denominators apply. Equal totals may conceal different regressions. The original clock remains primary.

S7.3 Earlier cohorts are not pooled

The earlier 30-condition study had source-copy 2, clipping 24, old geometry 28 and Full 24/27/22. Learning did not stably beat clipping, and all seeds trailed old geometry. The later 15-condition supplement had Full 9/11/9 versus Constant 7/6/8, including demo_179 seed 0: Full 0/5 versus Constant 5/5. These cohorts, FINAL and Lift remain separate; all clocks, exclusions and failures are retained. Expansion changes both data and update count, so it does not isolate either effect.

Complete Can outcomes: original clock

Registered sources		173					162					188					110					170					119				
Method	All /50	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2
Source copy	12	0	0	1	0	0	0	1	1	0	0	-	-	-	-	-	0	0	1	0	0	-	-	-	-	-	0	0	0	0	0
Clip / common	36	R	R	1	R	R	0	D	1	0	0	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	R	0	0
Clip / guard	36	R	R	1	R	R	0	D	1	0	0	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	R	0	0
Nearest / 36	37	R	R	1	R	R	0	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	0	0	0
Nearest / 59	33	0	0	1	0	0	0	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	0	0	0
Geometry / old	46	R	R	1	R	R	R	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	0	0
Geometry / position	47	R	R	1	R	R	R	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	R	0
Geometry / free R	43	R	R	1	R	R	R	1	1	R	R	-	-	-	-	-	0	0	1	0	0	-	-	-	-	-	R	R	R	0	R
Full / seed 0	45	R	R	1	R	R	R	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	R	R
Full / seed 1	43	R	R	1	R	R	R	1	1	R	R	-	-	-	-	-	0	R	1	R	R	-	-	-	-	-	R	0	R	0	0
Full / seed 2	43	R	R	1	R	R	R	1	1	R	R	-	-	-	-	-	0	R	1	R	R	-	-	-	-	-	R	R	0	R	R
Constant / seed 0	45	0	R	1	R	R	R	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	R	R
Constant / seed 1	37	0	R	1	0	0	R	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	R	R	0	R
Constant / seed 2	37	0	R	1	0	0	R	1	1	R	R	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	R	0	0

Registered sources		120					171					126					148					147					79				
Method	All /50	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2
Source copy	12	0	0	0	0	0	0	1	1	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	0	0	0	0	1	0	0
Clip / common	36	0	R	R	R	R	R	1	1	R	1	R	R	1	R	R	0	D	1	0	D	R	R	R	0	R	R	R	R	1	R	R
Clip / guard	36	0	R	R	R	R	R	1	1	R	1	R	R	1	R	R	0	D	1	0	D	R	R	R	0	R	R	R	R	1	R	R
Nearest / 36	37	0	0	R	0	0	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	R	0	0	R	R	R	1	R	R
Nearest / 59	33	0	0	R	0	0	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	R	0	0	R	R	R	1	R	R
Geometry / old	46	R	R	R	R	R	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	R	R	0	R	R	R	1	R	R
Geometry / position	47	0	R	R	R	R	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	R	R	R	R	R	R	1	R	R
Geometry / free R	43	R	R	R	R	R	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	R	R	0	R	R	R	1	R	R
Full / seed 0	45	R	R	R	R	R	R	1	1	R	1	R	R	1	R	R	0	1	1	R	1	0	R	0	0	R	R	0	1	R	R
Full / seed 1	43	R	R	R	R	R	R	1	1	R	1	R	R	1	R	R	R	1	1	R	D	0	R	R	0	R	R	R	1	R	R
Full / seed 2	43	R	R	R	R	R	R	1	1	R	1	R	R	1	R	R	0	1	1	0	1	0	R	0	0	R	R	R	1	R	R
Constant / seed 0	45	0	R	R	0	R	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	R	R	0	R	R	R	1	R	R
Constant / seed 1	37	0	0	R	0	0	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	0	R	0	0	R	R	1	R	R
Constant / seed 2	37	0	R	R	0	R	R	1	1	R	1	R	R	1	R	R	R	1	1	R	1	0	0	R	0	0	R	R	1	R	R

1 preserved success R repair 0 failure D success regression - excluded / NA

T / M / L: fitted ROMs; I1 / I2: withheld ROMs. Repeated totals cover all 50 scored conditions.

Figure S1. Complete Can matrix at the original, primary clock. Two stacked panels divide only the registered source columns; no method, seed or outcome is selected out. Each row's repeated total covers all 50 scored conditions, not just its panel. Across both panels, 700 cells are scored and 140 are normal-qualification NA. Codes distinguish retained successes (1), repairs (R), failures (0), success regressions (D) and exclusions (-). Every cell and all fourteen totals match the previous full-width matrix.

Reading the matrix. Compare each method with source-copy within the same source/ROM column. Full seed 1's regression remains visible; the geometry and retrieval results are retained as complete comparators. This display provides membership detail beyond main Table 1, not additional experiments.

Complete Can outcomes: +40 clock

Registered sources		173					162					188					110					170					119				
Method	All /50	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2
Source copy	18	0	0	1	0	0	0	1	1	1	1	-	-	-	-	-	0	0	1	0	0	-	-	-	-	-	0	0	0	0	0
Clip / common	36	R	R	1	R	R	0	D	1	D	D	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	R	0	0
Clip / guard	36	R	R	1	R	R	0	D	1	D	D	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	R	0	0
Nearest / 36	37	R	R	1	R	R	0	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	0	0	0
Nearest / 59	33	0	0	1	0	0	0	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	0	0	0
Geometry / old	46	R	R	1	R	R	R	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	0	0
Geometry / position	47	R	R	1	R	R	R	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	R	0
Geometry / free R	43	R	R	1	R	R	R	1	1	1	1	-	-	-	-	-	0	0	1	0	0	-	-	-	-	-	R	R	R	0	R
Full / seed 0	45	R	R	1	R	R	R	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	R	R
Full / seed 1	43	R	R	1	R	R	R	1	1	1	1	-	-	-	-	-	0	R	1	R	R	-	-	-	-	-	R	0	R	0	0
Full / seed 2	44	R	R	1	R	R	R	1	1	1	1	-	-	-	-	-	0	R	1	R	R	-	-	-	-	-	R	R	0	R	R
Constant / seed 0	45	0	R	1	R	R	R	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	R	R	R	R	R
Constant / seed 1	38	0	R	1	0	0	R	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	R	R	0	R
Constant / seed 2	41	0	R	1	R	R	R	1	1	1	1	-	-	-	-	-	R	R	1	R	R	-	-	-	-	-	0	0	R	0	0

Registered sources		120					171					126					148					147					79				
Method	All /50	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2	T	M	L	I1	I2
Source copy	18	0	0	1	0	0	0	1	1	1	1	0	0	1	0	0	0	1	1	0	1	0	0	0	0	0	0	1	1	0	1
Clip / common	36	0	R	1	R	R	R	1	1	1	1	R	R	1	R	R	0	D	1	0	D	R	R	R	0	R	R	1	1	R	1
Clip / guard	36	0	R	1	R	R	R	1	1	1	1	R	R	1	R	R	0	D	1	0	D	R	R	R	0	R	R	1	1	R	1
Nearest / 36	37	0	0	1	0	0	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	R	0	0	R	R	1	1	R	1
Nearest / 59	33	0	0	1	0	0	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	R	0	0	R	R	1	1	R	1
Geometry / old	46	R	R	1	R	R	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	R	R	0	R	R	1	1	R	1
Geometry / position	47	0	R	1	R	R	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	R	R	R	R	R	1	1	R	1
Geometry / free R	43	R	R	1	R	R	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	R	R	0	R	R	1	1	R	1
Full / seed 0	45	R	R	1	R	R	R	1	1	1	1	R	R	1	R	R	0	1	1	R	1	0	R	0	0	R	R	D	1	R	1
Full / seed 1	43	R	R	1	R	R	R	1	1	1	1	R	R	1	R	R	R	1	1	R	D	0	R	R	0	R	R	1	1	R	1
Full / seed 2	44	R	R	1	R	R	R	1	1	1	1	R	R	1	R	R	0	1	1	0	1	0	R	R	0	R	R	1	1	R	1
Constant / seed 0	45	0	R	1	0	R	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	R	R	0	R	R	1	1	R	1
Constant / seed 1	38	0	R	1	0	0	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	0	R	0	0	R	1	1	R	1
Constant / seed 2	41	0	R	1	0	R	R	1	1	1	1	R	R	1	R	R	R	1	1	R	1	0	R	R	0	R	R	1	1	R	1

1 preserved success R repair 0 failure D success regression - excluded / NA

T / M / L: fitted ROMs; I1 / I2: withheld ROMs. Repeated totals cover all 50 scored conditions.

Figure S2. Complete Can matrix at the separate +40 endpoint. Source order, methods, symbols and exclusions match Fig. S1, but repair/regression is evaluated against source-copy at this endpoint. Its 18 successes and 32 failures are not interchangeable with the primary 12/38 split. Full seed totals are 45/43/44; Constant totals are 45/38/41. All 840 displayed cells and fourteen totals match the archived +40 matrix. This clock does not replace the original endpoint.

Endpoint sensitivity. For example, Full seed 0 has 45 successes at both endpoints, yet its source-success regression count changes from zero to one. Equal totals are therefore not evidence of identical repair membership. The paired clock table and matrices retain these changes explicitly.

S8 Can Generation-Cost Scope

Main Fig. 2a already shows all fourteen Can rows, medians and condition IQRs, so that chart is not duplicated here. Timing covers features, gates, transfers, materialization, shape/finiteness checks and solver-record capture, with traces, models, banks and privileged FK resident. Setup, loading, serialization, qualification, teacher/training, execution and audit are excluded. All 700 condition medians are retained, failures included. IQRs are not confidence intervals. Neural rows add an RTX 5090 to the pinned CPU; these are not matched-resource planner speedups, online latency or training payback.

S9 Source-Disjoint Lift Supplement

Lift uses the registered robomimic proficient-human low-dimensional source and the same core simulator/controller versions and strict criterion. Source roles were frozen before target outcomes. Of 16 TRAIN nominees, 15 qualified, giving 45 conditions: 40 accepted identities and five source failures. All three fixed geometry candidates were executed for each failure: 13/15 passed. The first accepted candidate in old/position/free order supplied each repair teacher. No failed target received an invented positive label.

The same architecture was freshly initialized at seeds 0–2, not loaded from Can. Each CPU fit used AdamW, batch 8, learning rate 3×10^{-4}, weight decay 10^{-4}, balanced composed imitation and 5,000 updates. The checkpoint rule was fixed before TEST. Four DEV sources qualified; only prespecified demo_134/tight was used for the five-program interface check: source-copy and all learners failed, old geometry passed. Neither checkpoints nor seeds were changed.

TEST used the first ten registered sources, 69/84/54/183/18/71/186/99/166/127, with three ROMs. All qualified; all 300 programs were sealed before restricted execution. No source, method, seed, ROM or clock was replaced after an outcome. Full execution used 663,750 native steps and independent checks 690,300 calls, including the separately scored +40 endpoint.

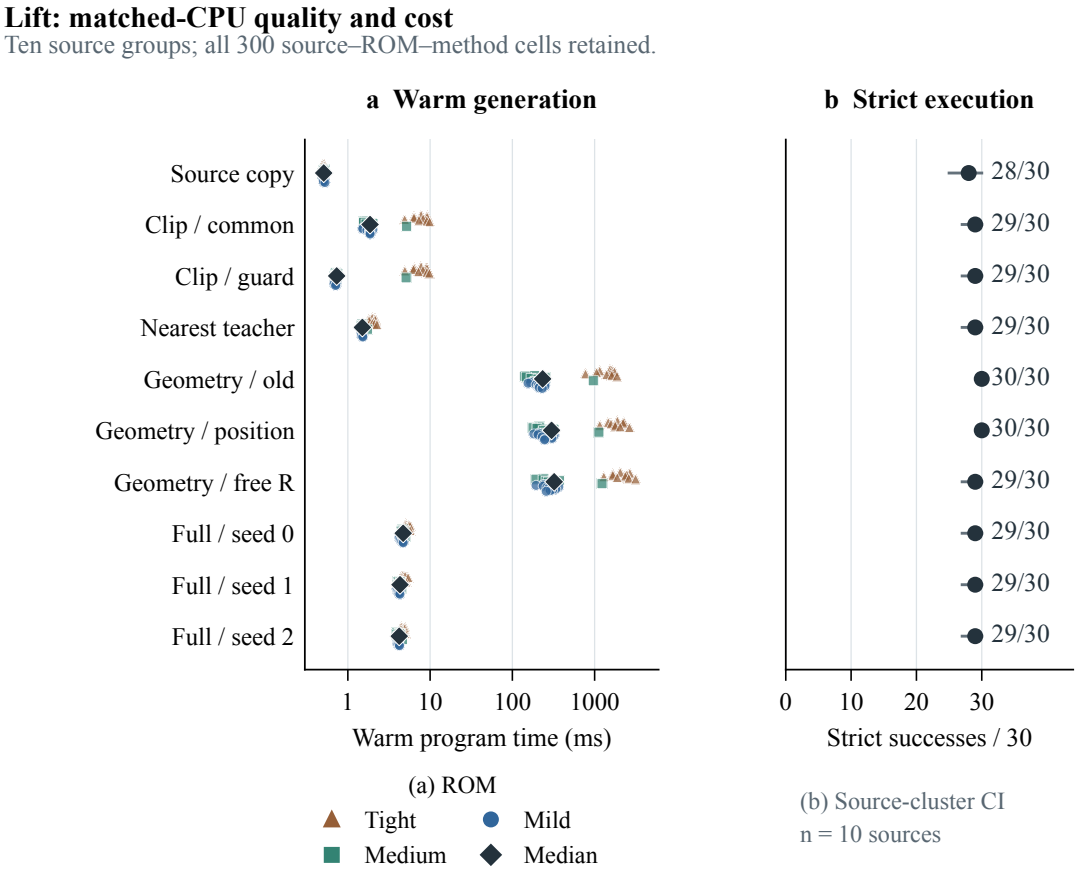


Figure S3. Lift's matched-CPU detail beyond main Fig. 2b. Left: every context median (three calls), ROM-specific shapes/colors, and each method's median (black diamond); deterministic offsets separate overlaps. Right: strict success and descriptive whole-source 95% intervals. Seeds are separate methods, not extra independent sources. The log-time axis describes warm generation only. Every context/method is retained.

S9.1 Failures and intervention

Every method passed the native goal in 30/30 conditions, which is not strict success. Source 127/tight held 7/8 carry actions with legal joints; nearest and every learner repaired it. Source 183/tight held 8/9 and exceeded finger 2 by 0.172 mm; nearest and all learners retained this failure. Old/position geometry repaired both with no observed regression; clipping/free geometry repaired both but regressed 99/medium.

Both source failures share bit-exact initial states with their normal executions. Guard intervention and state/motor change first occur at physical step 200; neither has an arm-range violation. Eleven of 30 source conditions record intervention/control or state changes; 19 show none in the checked original-clock fields. Both failures are among the eleven, but contact/carry and finger effects remain alternative explanations. All 300 original/+40 strict classifications match.

S9.2 Same-CPU generation and quality

All methods use one pinned CPU, single-thread libraries and CUDA invisible. A later read-only inspection identified Xeon Gold 6538Y+ and the original receipt's boot ID; it is not a new timing run. Each context/method has one discarded warm-up and three measured calls, with rotating method order: 1,200 calls, 900 timed. The 300 four-call output groups match, and each executed program matches its first measured output. Timing includes features, masks, generation, materialization and the unchanged contract; resident inputs/FK, exclusions and warm-cost limits follow S8. Unlike Can's historical GPU outputs, these scored CPU outputs determine Lift quality.

S9.3 Source uncertainty and the complete matrix

The descriptive plan predates restricted TEST execution. It resamples ten whole source groups with all ROMs/methods/timing (10,000 draws; seed 20261010). Strict intervals use all draws; repair intervals use 8,922 defined draws and retain 1,078 zero-denominator draws as NA. Models are fixed. Two source failures, source-copy 28/30 and degenerate intervals limit inference; zero observed regression does not establish zero unseen risk. Nearest matches every learner's outcome membership at lower cost. No bootstrap was repeated for this revision.

Registered sources		**69**			**84**			**54**			**183**			**18**			**71**			**186**			**99**			**166**			**127**		
Method	All /30	T	M	L	T	M	L	T	M	L	T	M	L	T	M	L	T	M	L	T	M	L	T	M	L	T	M	L	T	M	L
Source copy	28	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	x	1	1
Clip / common	29	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1
Clip / guard	29	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1
Nearest teacher	29	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Geometry / old	30	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Geometry / position	30	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Geometry / free R	29	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1
Full / seed 0	29	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Full / seed 1	29	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Full / seed 2	29	1	1	1	1	1	1	1	1	1	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

1 strict success x strict failure T / M / L: tight / medium / mild. All native goals pass.

Figure S4. Complete Lift matrix: ten sources, three ROMs and ten methods, with all 300 outcomes. The transposed layout retains every method/condition: 1 denotes success and x failure, despite native-goal success in every cell. The source failures at 183/tight and 127/tight and the regressions at 99/medium remain visible. Equal 29/30 totals can conceal different repair/regression membership. Both clocks have identical strict members; the repair denominator remains two.

S10 Material Access and Reference Stack

Author-maintained materials: https://github.com/aownown/wrist-range-repair (not SPIE-hosted). The companion supplies 840 Can declared rows, 300 Lift scored rows, timing/uncertainty, source/teacher roles, DEV history and method listings; it excludes dataset tensors, weights and full dense trajectories. Separately, the review archive retains six Can/three Lift weights, teacher banks, selected source/model/assets, code, execution records and publication sources with hashes. Complete TRAIN dynamic closure and full-study runtime reproduction remain unestablished. Third-party materials retain their original terms.

The observed stack is Python 3.10.12, NumPy 1.26.4, SciPy 1.15.3, MuJoCo 2.3.2, robosuite 1.4.1 and PyTorch 2.7.1+cu128; the wheel tag does not certify drivers/toolkits. Shared-storage retention is not an off-site backup, and static checks do not resolve the failed CPU numerical replay. Lift is independently trained, not zero-shot, visual, cross-embodiment or hardware evidence.

S11 Numerical Portability and Finite GPU Reproduction

S11.1 CPU negative result retained

The first frozen TRAIN case, demo_184/tight (seed 0, update 2,000; 130 actions), failed all four numerical checks. At atol = rtol = 10^{-5}, raw rotation/reference maxima were 1.6911×10^{-5} and 2.5171×10^{-4}, with 47/460 entries outside tolerance. Materialized rotation/reference maxima were 2.1728×10^{-5} and 8.9622×10^{-5} rad, both exceeding the fixed 2×10^{-5} absolute limits.

One forward ran, without physics/training; XYZ, gripper, clock, homogeneous rows and 53 zero-gate rows stayed intact. Execution impact/backend cause remain unknown. The later frozen-GPU-output materialization check also failed; neither negative is resolved by the independent Lift CPU study.

S11.2 Bounded same-host GPU check

On the original RTX 5090 host, a fresh isolated runtime matched S10's core versions. Three frozen forwards (first TRAIN case; two preselected FINAL cases, seed 0/update 5,000) passed independent NumPy/SciPy input/gate/clock checks, with bit-equal programs and zero latent/materialization error. No training ran. Only then did the two FINAL programs execute continuously: 7,325 integrator and 7,618 independent audit calls. Both-clock outcomes are in Table S3.

Table S3. Finite reference-GPU execution. These preselected cases are not new generalization trials; carry is held/required actions.

Case	Actions	Carry	Strict original/+40	All-nine legal	Illegal joint
119/tight	120	53/53	yes/yes	yes	none
79/medium	93	39/39	no/no	no	J7

The 113-file record is not an installed environment or full Can/TRAIN replay. Cross-host/fresh-OS replication, new timing and learner necessity are not established. The CPU result remains accepted=false.

S12 Source Heterogeneity and Recorded Scenes

Figure S5 retains complete repair composition and source differences for the same cohort; see S7.1.

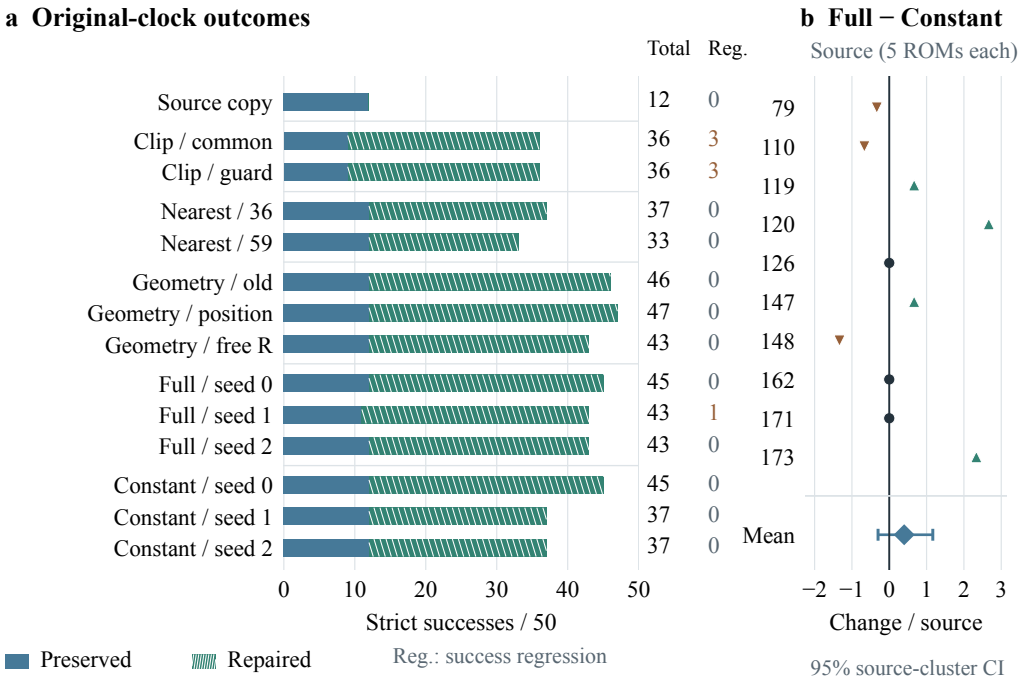


Figure S5. Can heterogeneity. All fourteen methods/seeds retain success/repair/regression composition (left); all ten paired source differences are shown (right). Each difference sums five ROMs and averages three seeds. Mean: 0.4 successes per source (8 percentage points); the exploratory source-cluster interval crosses zero.

S12.1 Matched scenes and full telemetry

The same archived runs as main Fig. 3 retain all five rows, three original times and unchanged crops. These outcome-selected illustrations do not estimate prevalence; matching commands/clocks do not imply matching measured states, forces or contacts.

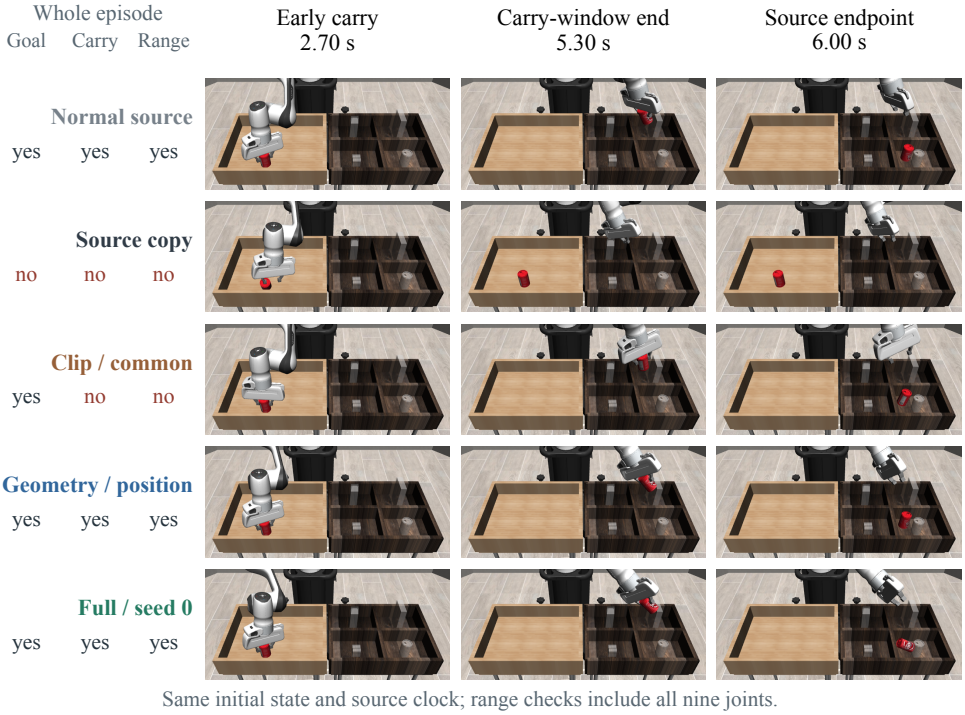


Figure S6. Complete demo119/tight scene record: all fifteen frames and goal/carry/range checks. Geometry is position-priority. Source-copy violates a finger limit, not proof of direct J7 infeasibility. Frames illustrate, not establish, continuous success.

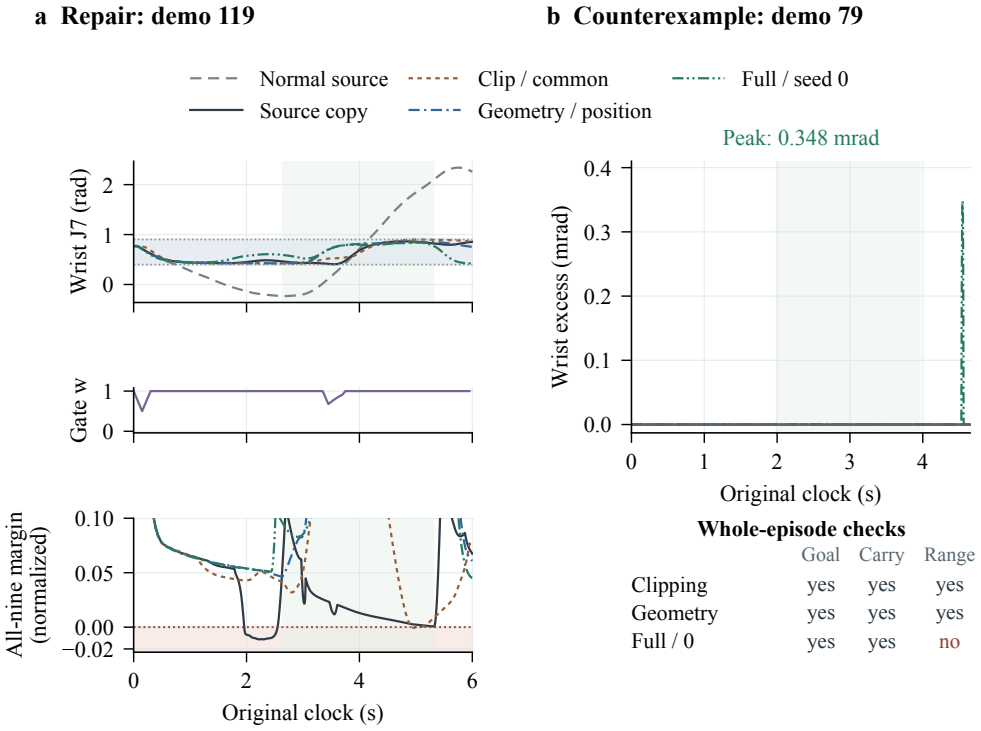


Figure S7. Full J7, gate and all-nine telemetry. Demo79 Full seed 0 exceeds J7 by 0.348 mrad despite goal/carry success; clipping/geometry pass. Normal traces use nominal bounds, restricted rows target bounds. Original carry shading and near-zero margin zoom remain; no smoothing or rerun.

S12.2 A retained source success lost after editing

Main Section 5.3 and Fig. S1 retain Full seed 1's regression on demo148/I2. Figure S8 expands that archived case with all five recorded comparison rows. It illustrates a failure mode, not its population frequency or a new experiment.

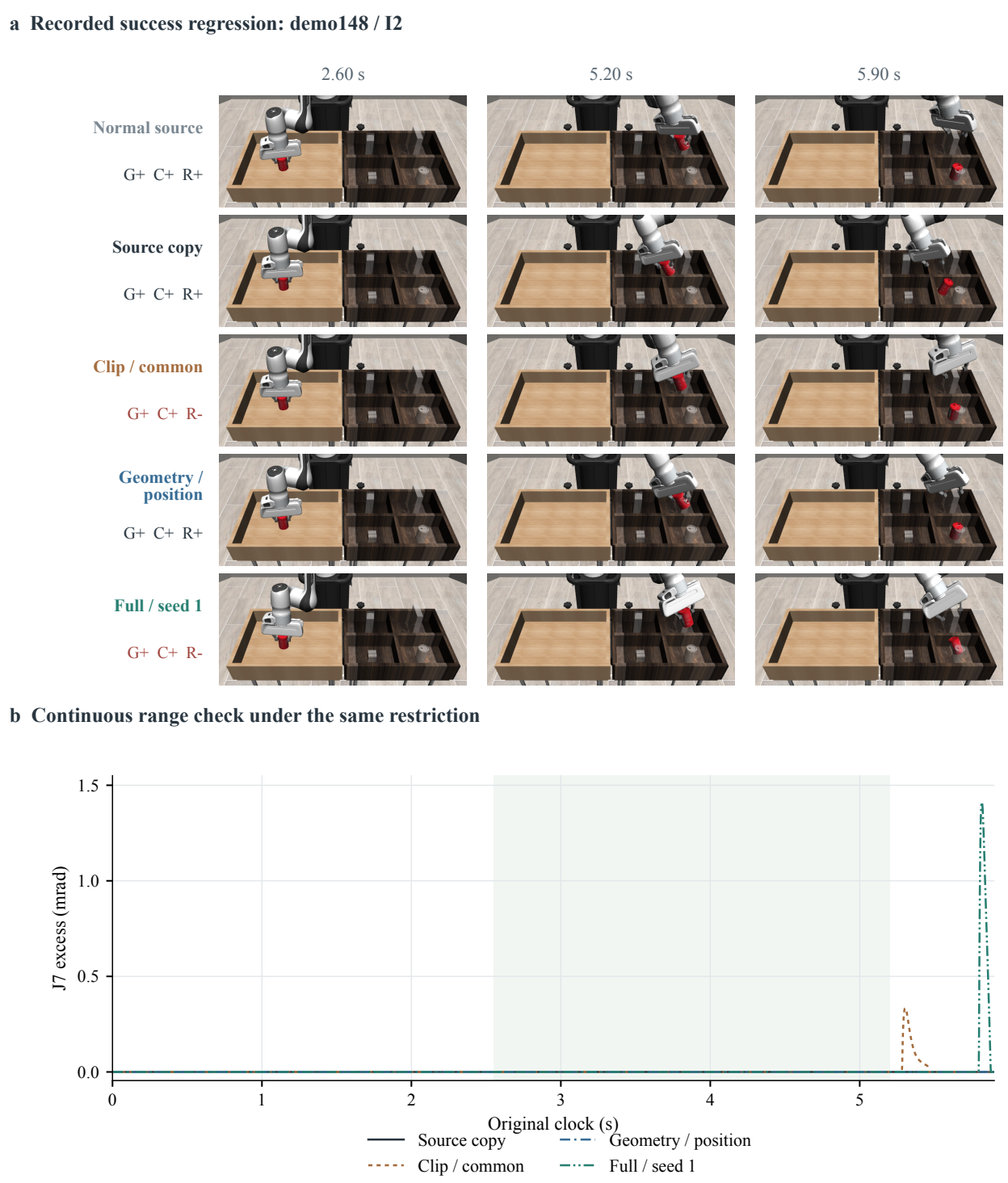


Figure S8. Recorded success regression, demo148 at the second withheld ROM. All five rows share initial-state bytes and original times. Goal/carry pass throughout, but clipping and Full seed 1 exceed J7 by 0.333 and 1.410 mrad; source-copy and position geometry remain strict successes. All four restricted J7 traces are shown; source-copy and geometry overlap at zero. Frames retain the common crop; continuous checks, not images, determine outcomes.

S13 DEV Selection and the Compact Companion

S13.1 Two-stage family/update selection

Stage 1 selects a shared Full update from the original 36-teacher family; stage 2 compares that selected family with expanded-59/5,000 (Table S4). The rule maximizes equal-source/equal-seed original-clock Full strict success, then prefers fewer source-success regressions and earlier updates. DEV has no restricted source positives, so regression is NA. Constant, +40, fitting loss and FINAL do not choose the model or seed. All Full/Constant models share the chosen family/update.

Table S4. Complete recorded Can DEV history. Each seed uses the same 15 conditions from five sources; totals /45 are correlated seed–condition counts. Stage 2 repeats the selected stage 1 comparator.

Stage	Teachers / updates	Seed 0	Seed 1	Seed 2	Total /45
1	**36 / 2,000**	14	8	8	**30**
1	36 / 5,000	11	6	9	26
1	36 / 10,000	9	8	9	26
2	36 / 2,000	14	8	8	30
2	**59 / 5,000**	11	10	10	**31**

Constant's descriptive 36/2,000 and 59/5,000 counts are 10/7/10 and 9/12/8. Full gains only one of 45, with eight gains and seven regressions; this is not a significant-gain claim, and data/update effects remain inseparable.

S13.2 What the companion verifies

evidence_companion/verify_tables.py re-aggregates stored successes, repairs/regressions, medians and source differences, checks source-role separation and included hashes. It performs no inference, physics, FK, new bootstrap or external-file verification. Inspection listings are not an installable runtime; full dense trajectories, original TRAIN prequalification nominations and complete dynamic reproduction remain outside the compact package.

The figure catalogue records locations, purposes and source data. Main Fig. 2 shows generation costs and paired Can source differences; main Tables 1–2 retain strict outcome totals. Complete outcome matrices and extended scene and telemetry records are provided in this supplement.

Verification note. Existing outcomes, seeds, two clocks, exclusions, tolerances and negative reproduction findings are unchanged. Authors remain responsible for verification and final approval. Public availability does not establish complete dynamic reproducibility or grant rights to excluded third-party assets.